

SLURRY_FREE_ACID Modeling

Model Upgrade Addendum

Work completed after the first project advancement report

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Main result: the target-free Ridge-LightGBM model was improved retrospectively and a regime-aware hybrid reduced forward-validation fragility. The two candidates have different strengths; neither is externally validated because only the January month is available.

Executive summary

After the first report, I focused on improving the **target-free virtual sensor**: process measurements are used to estimate the current free-acid value without using current or historical `SLURRY_FREE_ACID` as an input.

I completed two additional notebooks:

- **Notebook 06**: compared recency weighting, clustered local experts, stronger Ridge regularization, LightGBM alternatives, and Ridge–LightGBM blends using four expanding validation blocks;
- **Notebook 07**: combined the global blend with five local Ridge experts to test whether operating-region specialization improves robustness.

Candidate	Mean validation RMSE	Worst-block RMSE	Decision
Notebook 06 global 50/50 blend	0.3201	0.4124	Accuracy-oriented candidate; retrospective test RMSE 0.2879.
Notebook 07 80% local hybrid	0.3217	0.3520	Robustness-oriented candidate; retrospective test RMSE 0.2889.

Decision: keep both candidates frozen. Notebook 06 is slightly better on the already-inspected test period. Notebook 07 is more stable across forward validation blocks. A genuinely later labeled month is required to choose between them.

1 Scope retained from the first report

The original experimental design was not changed:

- the preparation layer and Feature Set D are inherited from Notebook 02;
- timestamps remain chronological and data are never shuffled;
- preprocessing is fitted inside each training fold only;
- the target, future-target columns, and target history are absent from the model inputs;
- January’s final test period is reported as **retrospective**, because it was already inspected previously.

The three operational cases from the first report remain separate:

- direct future-level forecasting;
- target-anchored change forecasting when y_t is available;
- target-free current-value estimation when no free-acid measurements are supplied.

The new work concerns only the third case.

2 Notebook 06: controlled target-free upgrades

Question

Could a compact adaptation method improve the process-only virtual sensor without introducing target leakage or unnecessary sequence-model complexity?

Models tested

Method	Purpose	Executed choices
Global Ridge	Stable regularized linear reference	$\alpha = 10$ and 100; median imputation and standard scaling
Recency-weighted Ridge	Give recent process states more influence	Half-lives of 3, 7, and 14 days
Clustered Ridge experts	Fit separate linear relationships in data-defined operating regions	KMeans with 3 or 5 clusters; one Ridge expert per cluster
LightGBM	Learn nonlinear thresholds and interactions	150 trees; 15 or 31 leaves; learning rate 0.05
Ridge–LightGBM blend	Combine linear stability and nonlinear flexibility	Ridge weights 0.3, 0.5, and 0.7

Validation design

The original validation interval was divided into four consecutive blocks. For each block, the training set contained the original training data plus only the earlier validation blocks.

Block	Training rows	Validation rows	Validation period
1	30,940	1,657	22 Jan 17:18 to 23 Jan 21:08
2	32,597	1,657	23 Jan 21:09 to 25 Jan 00:49
3	34,254	1,657	25 Jan 00:50 to 26 Jan 04:30
4	35,911	1,657	26 Jan 04:31 to 27 Jan 08:58

Selection was based on candidates within 2% of the best mean RMSE, followed by the lowest worst-block RMSE and RMSE standard deviation. The test period was not used for this decision.

Executed results

Candidate	Mean RMSE	RMSE SD	Worst RMSE
70% Ridge / 30% LightGBM	0.3191	0.0762	0.4236
50% Ridge / 50% LightGBM	0.3201	0.0686	0.4124
Five-cluster Ridge	0.3280	0.0396	0.3522
Global Ridge, $\alpha = 100$	0.3274	0.0811	0.4353
LightGBM, 31 leaves	0.3483	0.0599	0.4080

The selected candidate was the 50/50 Ridge–LightGBM blend. Compared with the Notebook 05 frozen blend, refitting on training plus validation reduced retrospective test RMSE from 0.3014 to 0.2879 and MAE from 0.2490 to 0.2344. This is a 4.48% RMSE reduction, but it is not fresh external evidence.

Interpretation

- Recency weighting did not improve performance; simple exponential down-weighting was not sufficient.
- LightGBM alone was weaker than the blend.
- The five-cluster Ridge model was more stable, which motivated Notebook 07.
- The retrospective gain mainly reflects stronger validation and refitting with more development data, not a completely new architecture.

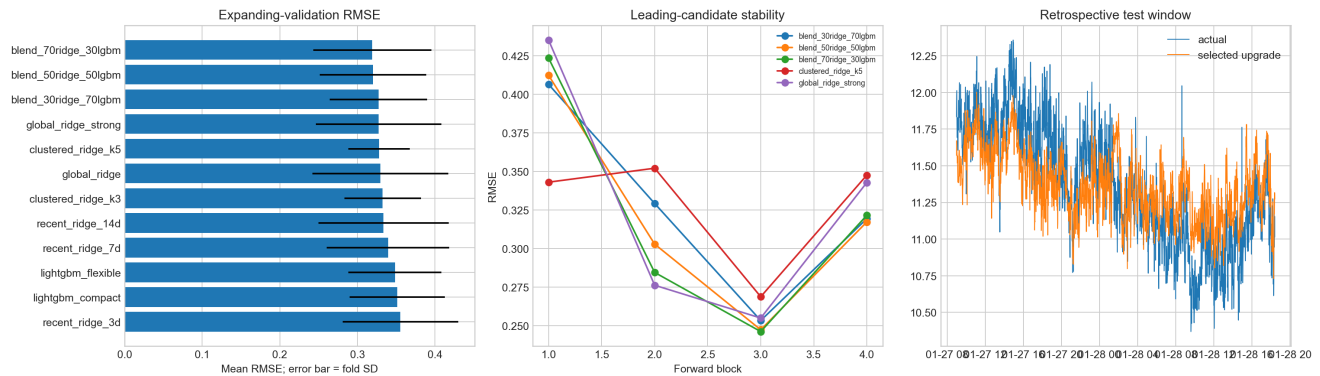


Figure 1: Notebook 06 benchmark. The left panel compares mean expanding-validation RMSE, the center shows time-block variability, and the right panel is the already-inspected retrospective test window.

3 Notebook 07: regime-aware hybrid

Question and implementation

Notebook 06 showed a trade-off: the global blend had better average accuracy, while local Ridge experts had a better worst block. I therefore combined both components.

For each validation fold:

- median imputation and standard scaling were fitted on the fold's training rows;
- the global prediction was

$$\hat{y}_{\text{global}} = 0.5\hat{y}_{\text{Ridge}} + 0.5\hat{y}_{\text{LightGBM}};$$

- KMeans divided the standardized training rows into five data-defined regions;
- one Ridge expert was fitted per region, with a global Ridge fallback for regions containing fewer than 100 training rows;
- fixed local weights from 0.0 to 1.0, in steps of 0.1, were evaluated;
- distance gates at the 50th, 70th, 85th, and 95th percentiles were also evaluated.

The fixed hybrid is

$$\hat{y} = (1 - w)\hat{y}_{\text{global}} + w\hat{y}_{\text{local}}.$$

KMeans clusters are statistical operating regions, not confirmed production regimes.

Main validation results

Local weight	Mean RMSE	RMSE SD	Worst RMSE
0.0 - global only	0.3201	0.0686	0.4124
0.4 - best mean	0.3160	0.0522	0.3777
0.5	0.3165	0.0487	0.3703
0.8 - selected	0.3217	0.0412	0.3520
1.0 - local only	0.3280	0.0396	0.3522

The 40% local blend had the best mean RMSE. The stability-first rule selected the 80% local blend because its mean remained within 2% of the best and it had the lowest worst-block RMSE in that pool.

Relative to the global-only model, the selected hybrid:

- changed mean RMSE from 0.3201 to 0.3217: approximately 0.5% worse;
- reduced worst-block RMSE from 0.4124 to 0.3520: approximately 14.7% better;
- reduced RMSE standard deviation from 0.0686 to 0.0412: approximately 39.9% lower;
- reduced mean MAE from 0.2546 to 0.2500.

Retrospective comparison

Candidate	Test MAE	Test RMSE	Test R^2
Notebook 06 global blend	0.2344	0.2879	0.5970
Notebook 07 selected hybrid	0.2309	0.2889	0.5941

The hybrid improved MAE slightly but increased RMSE by 0.36%. This indicates a different error distribution, not a clear overall win.

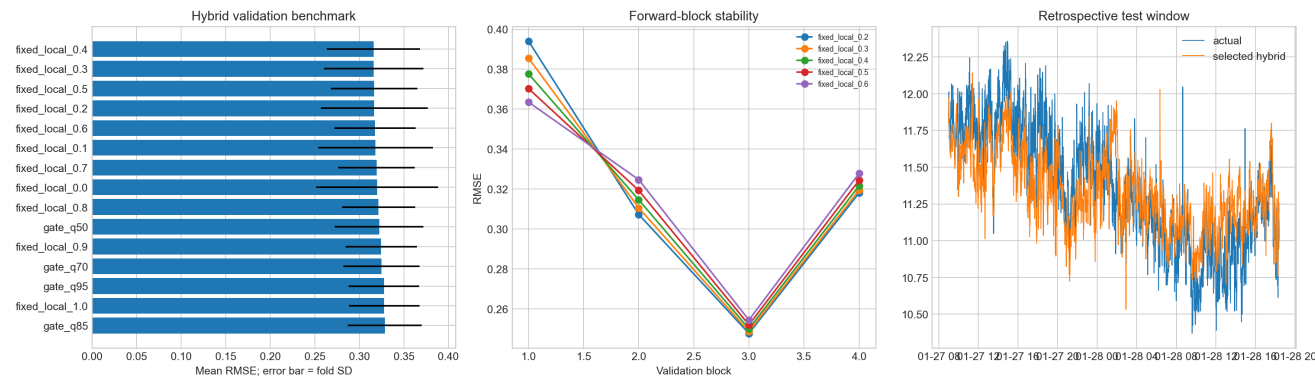


Figure 2: Notebook 07 results. Fixed blends gave a better mean/stability trade-off than hard distance gates. The test trace is retrospective and is not used for selection.

4 Engineering interpretation and limitations

- **Average accuracy and robustness are different objectives.** The best mean model is not the model with the best worst-period performance.
- **Local structure is present, but not yet physically labeled.** Some KMeans regions had no rows in particular validation blocks, showing that operating coverage changes with time.
- **Hard distance gating was not superior.** Smooth fixed blending was more reliable than switching abruptly between global and local predictions.
- **Only one month is available.** Adjacent one-minute samples are not equivalent to independent months or production campaigns.
- **The January test is no longer untouched.** It cannot support a final model replacement claim.
- **No causal or control recommendation follows from these models.** The variable registry and process-engineer confirmation remain mandatory.

5 Final model status

Role	Current candidate	Reason
Accuracy-oriented target-free model	Notebook 06 50/50 Ridge-LightGBM	Lowest retrospective RMSE and stronger R^2
Robustness-oriented target-free model	Notebook 07 80% local hybrid	Much lower worst-block RMSE and variability
Mean-validation alternative	Notebook 07 40% local hybrid	Lowest mean expanding-validation RMSE
Operational model	Not selected	Requires a genuinely later labeled month

All Notebook 07 readiness checks passed:

- no target or target-history input;
- four chronological forward blocks and no shuffle;

- preprocessing fitted inside each fold;
- test excluded from candidate selection;
- model bundle, predictions, tables, diagnostics, and readiness files exported.

6 Next work

1. Obtain at least one later month containing process variables and measured `SLURRY_FREE_ACID`.
2. Freeze Notebook 06 and Notebook 07 candidates before loading that month.
3. Compare RMSE, MAE, bias, drift, transition errors, and missing-sensor behavior without retuning.
4. Ask the process team whether the KMeans regions correspond to known grades, production rates, startups, shutdowns, or equipment states.
5. Select one operational candidate only if its improvement is consistent across months and practically meaningful.
6. Recalibrate prediction intervals and dashboard drift thresholds on multi-month data.

Deep sequence models should remain deferred until several chronological months are available. With the current data, external generalization and regime coverage are more important than adding model complexity.

Addendum change log

This addendum adds the work not present in the first advancement report:

- expanding four-block model validation;
- recency-weighted and clustered target-free alternatives;
- the revised Ridge-LightGBM retrospective result;
- fixed and distance-gated global/local hybrids;
- the explicit accuracy-versus-robustness decision;
- the requirement for a new external month before model replacement.